

A Non-Paninian Approach to Sanskrit Morphophonology

Ivan Tolchelnikov

Moscow State University

Sanskrit Zealots' Society

SRI Auroville

19th WSC, 2025

Outline

1. Vague Rules Issue in the European Approaches towards Accurate Description of Sanskrit Morphophonology;
2. Means to its Solution: Zalizniak's Concepts;
3. Summary and Achievements.

Vague Rules Issue in the European Approaches

Vague Rules Issue

Goal: a grammar terminologically approximate to European linguistics that generate correct wordforms:

- sufficiently accurate for computer use;
- generalizing corpus data (generation $\Rightarrow$ corpora);
- via algorithms;
- with minimal overgeneration.

Future Stem in Whitney (1)

Whitney W.D., Sanskrit Grammar (1896):

§932. [In the future tense] the root has the **guṇa**-strengthening.

√ <i>ji</i> “to win”	+	-sya-	=	<i>jesya-</i>	guṇa
√ <i>kṛ</i> “to make”	+	-sya-	=	<i>kaṛsya-</i> / <i>kaṛiṣya-</i>	guṇa
√ <i>bhū</i> “to be”	+	-sya-	=	<i>bhaviṣya-</i>	guṇa
√ <i>dhau</i> “to run”	+	-sya-	=	<i>dhāvṛddhiṣya-</i>	vṛddhi
√ <i>tai</i> “to stretch”	+	-sya-	=	<i>tāyṛddhiṣya-</i>	vṛddhi
√ <i>hiṃs</i> “to injure”	+	-sya-	=	<i>hiṃsiṣya-</i>	zero
√ <i>jṛmbh</i> “to gape”	+	-sya-	=	<i>jṛmbhiṣya-</i>	zero

Future Stem in Whitney (2)

Based on Whitney W.D., The Roots, Verb-Forms and Primary Derivatives of the Sanskrit Language (1885), we calculated:

815 verbal roots (few not stated in Whitney, e.g., √kal ‘to count’):

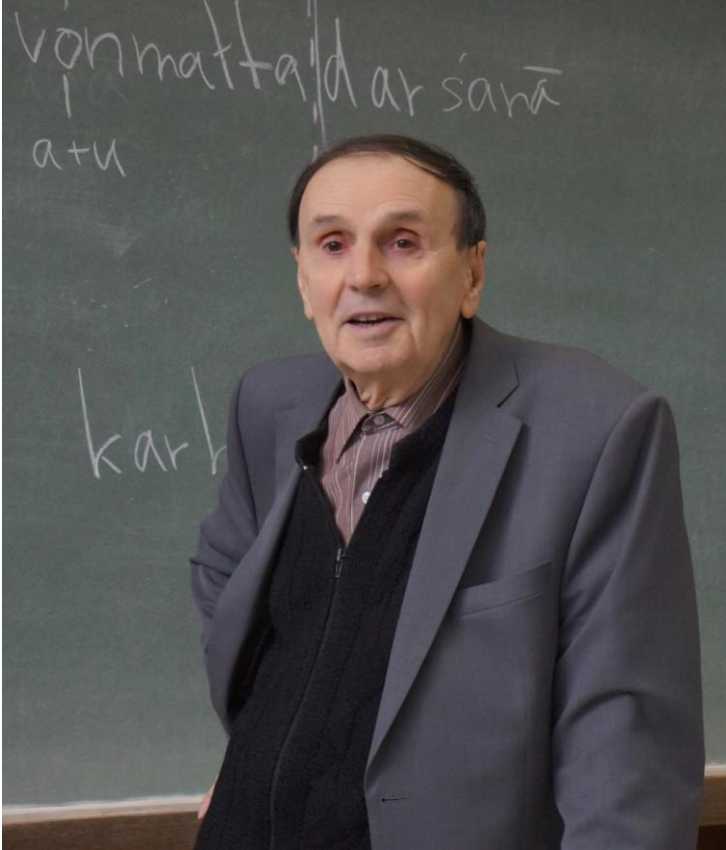
- not doubtful or artificial;
- can form future tense (unlike √as ‘to be’);
- root-variants (√dhmā//dham) were counted as separate.

109 do not have guṇa grade in future stem → 109 exceptions in the rule;

$$\frac{815-109}{815} \times 100\% \approx \mathbf{86.6\%} \text{ accuracy of the Whitney's rule §932.}$$

Means to Vague Rules Issue Solution: Zalizniak's Concepts

Andrei Zalizniak: Structuralism in Sanskritology



A. Zalizniak (1935–2017),
explaining Mahābhārata

“Morphophonological Classification of the
Sanskrit Language Verbal Roots”
(1975 Russian, 2025 English):

- Analysis of Whitney’s and Vaiyākaraṇas’ principles for verbal root nomenclature;
- **Original classification** for Sanskrit verbal roots regarding their alternation;
- Sketch of seṭ-veṭ-aniṭ parameters clarification.

Morphophonological Processes

How to transform a chain of morphemes $A+B+...+N$ into actual word $AB...N$, and backwards?

E.g., $\sqrt{yuj}$ 'to connect' + **caus.** **desid.** **ppp.** **n.** **N.** **sg.** $\rightarrow$
 $\sqrt{yuj} + -ay- + -s- + -ta- + -m \rightarrow ??? \rightarrow yuyojayi\dot{s}itam$

Processes	Example
Inserting connective $-i-/i-$	<i>yuyojayi\dot{s}itam</i>
Alternation	<i>yuyojayişitam</i>
Root reduplication	<i>yuyojayişitam</i>
Shift of the alternating element	E.g., atypical form of $\sqrt{sth\bar{a}}$: <i>stheya-</i>
Sandhi in verbal root alternating phoneme	E.g., $\sqrt{k\dot{r}} \rightarrow kriyate$ ($\dot{r}y \rightarrow riy$)
General sandhi	<i>yuyojayi\dot{s}itam</i>

Zalizniak's Alternation Series: Verbal Roots

Series	Zero/Weak	Guṇa/Middle	Vṛddhi/Lengthened
A ₁	∅s	as	ās
A ₂	sth∅	sthā	sthā
I ₁	ji	je	jai
I ₂	nī	ne	nai
U ₁	stu	sto	stau
U ₂	bhū	bho	bhau
R ₁	kṛ	kar	kār
R ₂	pṛ	par	pār
L	klp	kalp	kālp
M ₁	gṃ	gam	gām
M ₂	śṃ	śam	śām
N ₁	mṇ	man	mān
N ₂	jṇ	jan	jān

Zalizniak's Alternation Series: Application to Affixes

Series	Zero/Weak	Guṇa/Middle	Vṛddhi/Lengthened
A ₁	<i>yuñṣjanti</i>	<i>yunakti</i>	—
A ₂	<i>jāṇṣanti</i>	<i>jānāti</i>	—
I ₁	<i>buddhiḥ</i>	<i>buddheḥ</i>	<i>vairāgyam</i>
I ₂	<i>bhagavatiḥ</i>	—	—
U ₁	<i>krṇvanti</i>	<i>krṇoti</i>	<i>aupaniṣadaḥ</i>
U ₂	<i>tanūḥ</i>	—	—
R ₁	<i>kartṛbhiḥ</i>	<i>kartari</i>	<i>kartāram</i>
R ₂	—	—	—
L	—	—	—
M ₁	—	—	—
M ₂	—	—	—
N ₁	<i>rājṇā</i>	<i>he rājan</i>	<i>rājānam</i>
N ₂	—	—	—

Alternation: Morphological Position (1)

Cf.:

$\sqrt{i} + -ti \rightarrow \text{eti}$;

$\sqrt{i} + -anti \rightarrow \text{yanti}$;

$\sqrt{kr} + -nu- + -ti \rightarrow \text{kṛṇoti}$;

$\sqrt{kr} + -nu- + -anti \rightarrow \text{kṛṇvanti}$;

$\Rightarrow -ti$ makes preceding morpheme **guṇa**, $-anti$ – **zero**.

Morphological position – the alternation grade that a morpheme is **expected** to exhibit given a **certain immediately following morpheme**.

The expectation **may not be met**:

$\sqrt{as} + -anti \rightarrow \text{ṣanti}$;

$\sqrt{ad} + -anti \rightarrow \text{adanti}$, not $*\text{ṣdanti}$ as expected.

Alternation: Morphological Position (2)

Morphological position	Expected grade	Example
1st	zero	ppp. – (1) <i>ta-</i> : <i>kṛta-</i> , <i>ita-</i> BUT <i>carita-</i> , <i>sevita-</i> , <i>dhāvita-</i>
2nd	guṇa	future – (2) <i>śya-</i> : <i>kariṣya-</i> , <i>bhaviṣya-</i> BUT <i>jimbhiṣya-</i> , <i>dhāviṣya-</i>
3rd	vṛddhi	aorist 5 – (1/2/3) <i>iṣ-</i> : e.g., <i>vāriṣ-</i> , <i>cāyiṣ-</i>

Alternation Types

Alternation type – alternation pattern of a morpheme in three morphological positions.

MP Type	1st	2nd	3rd
	ppp. - (1) ta-	future - (2) sya-	aorist 5 - (3) is-
I $\sqrt{vr}$ (I)	<i>vr̥ta-</i>	<i>var̥iṣya-</i>	<i>vār̥iṣ-</i>
II $\sqrt{car}$ (II)	<i>car̥ita-</i>	<i>car̥iṣya-</i>	<i>cār̥iṣ-</i>
III $\sqrt{id}$ (III)	<i>īṭta-</i>	<i>īḍiṣya-</i>	<i>aiḍiṣ-</i>
IV $\sqrt{cai}$ (IV)	<i>cāyita-</i>	<i>cāyiṣya-</i>	<i>cāyiṣ-</i>

Alternation Types

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	ppp. - (1) ta-	future - (2) sya-	aorist 5 - (3) is-
I √vr̥ (I)	<i>vr̥ta-</i>	<i>vr̥ṣya-</i>	<i>vr̥ṣ-</i>
II √car (II)	<i>carita-</i>	<i>carṣya-</i>	<i>cārṣ-</i>
III √īḍ (III)	<i>īṭta-</i>	<i>īḍṣya-</i>	<i>aiḍṣ-</i>
IV √cai (IV)	<i>cāyita-</i>	<i>cāyṣya-</i>	<i>cāyṣ-</i>

Rule Issue Solution

Rule in Whitney:

§932. [In the future tense] the root has the **guṇa**-strengthening.

⇒ ≈ 109 exceptions, ≈ **86.6%** accuracy.

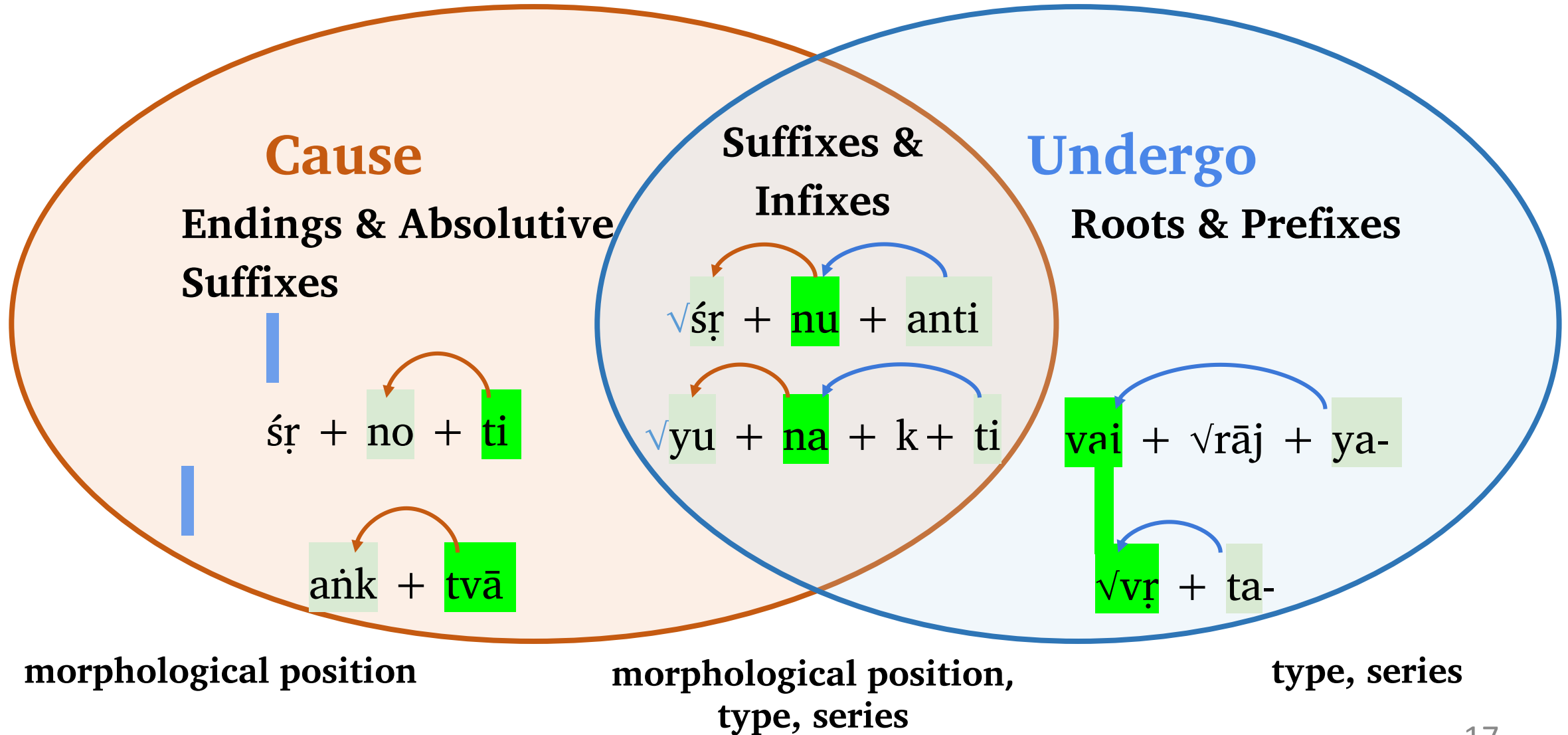
√jī (I) “to win”	+	-sya-	=	<i>jesya-</i>	guṇa
√car (II) “to move”	+	-sya-	=	<i>carīṣya-</i>	guṇa
√hiṃs (III) “to injure”	+	-sya-	=	<i>hiṃsiṣya-</i>	zero
√dhau (IV) “to run”	+	-sya-	=	<i>dhāvīṣya-</i>	vṛddhi

Rule with Zalizniak’s parameters: In the future tense the root is set in the **2nd morphological position**;

⇒ 1 exception (√mṛj ‘to wipe’ → *mārḱṣya-*, *mārjīṣya-*);

$\frac{815-1}{815} \times 100\% \approx \mathbf{99.9\%}$ accuracy.

Classification of Morphemes on the basis of Alternation



Summary and Achievements

Morphophonological Processes: Algorithm

How to transform a chain of morphemes $A+B+\dots+N$ into actual word $AB\dots N$ (from $AB\dots N$ to $A+B+\dots+N$ backwards)?

Step	Name	Comment
1	Inserting -i-/-ī-	General rule + “seṭ”-parameter for verbal roots
2	Alternation	Choice of the grade within alternation series acc. to morphological position and alternation type
3	Root reduplication	If required by suffix, determined by alternation series + phonological rules
4	Shift of the alternating element to another alternation series // Sandhi in verbal root	If required by following morpheme, described in terms of alternation series ($\sqrt{\text{sthā}}$, guṇa A_2 , $\rightarrow$ stheya-, guṇa I) // For these sandhi, see Zalizniak, 1975, Table 1
5	General sandhi	See Emenau 1968 + additional rules

Morphophonological Generator (alpha)

Perfect Tense									
P.					A.				
	sg.	du.	pl.		sg.	du.	pl.		
1	uvāca ✓ uvaca ✓	ūciva	ūcima ✓	1	ūce ✓	ūcivahe	ūcimahe		
2	uvacitha uvaktha	ūcathur	ūca	2	ūciṣe ✓	ūcāthe	ūgdhve		
3	uvāca ✓	ūcatur ✓	ūcus ✓	3	ūce ✓	ūcāte	ūcire		



Andrei Shirobokov

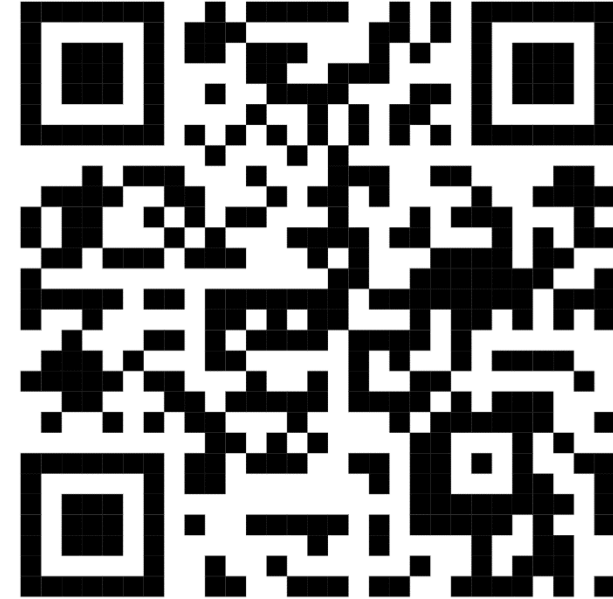
✓ – verified with
Sanskrit-Russian
Parallel Corpus

Thank you for attention!
vanishtax@gmail.com

Morphological generator
based on this approach
(A. Shirobokov)



“Morphophonological Classification of
the Sanskrit Language Verbal Roots”
A. Zalizniak (English)



Connective Vowel: Conditions

General rule: Connective vowel appears between morphemes A and B if:

- A doesn't end in $\bar{a}$ or zero grade;
- B starts with a consonant $\neq y, n$ (exc.: v, m) AND is not an aorist suffix.

Cases	-i-	-ī-
1. root + suffix	General rule + $seṭ/veṭ^*$	
2. root + ending	irregular appearance (<i>bravīti, roditi</i> , etc.)	
3. suffix + suffix	General rule	—
4. suffix + ending	—	Only in aorist: $-s-/ -siṣ-$ $/-sa-$ + 2–3.sg.P $-s/-t$

Connective Vowel: Set – Vet(s) - Anit

root		future	desiderative	ppp	else- where	number
set	<i>aṭ</i>	<i>aṭiṣya-</i>	<i>aṭitiṣ-</i>	<i>aṭita-</i>	set	≈ 320
anit	<i>pac</i>	<i>pakṣya-</i>	<i>pipakṣ-</i>	<i>pakta-</i>	anit	≈ 180
vet	<i>tap</i>	<i>tapīṣya-/tapsya-</i>	<i>ṭitapīṣ-/titaps-</i>	<i>tapīta-/tapta-</i>	vet	≈ 40
vet1	<i>takṣ</i>	<i>takṣīṣya-/takṣya</i>	<i>titakṣīṣ-/titakṣ-</i>	<i>taṣṭa-</i>	vet	≈ 100
vet2	<i>lap</i>	<i>lapīṣya-</i>	<i>lilapīṣ-</i>	<i>lapīta-/lapta-</i>	vet	≈ 60
vet3	<i>tras</i>	<i>trasiṣya-</i>	<i>titrasiṣ-</i>	<i>trasta-</i>	vet	≈ 70
vet4	<i>tyaj</i>	<i>tyajīṣya-/tyakṣya-</i>	<i>tityakṣ-</i>	<i>tyakta-</i>	vet	≈ 20